Name:

Quiz 1 Solutions

Instructions. The only tools you are permitted to use are pencils, pens, erasers, a handwritten sheet of notes, and your mind. No electronic devices! You have 1 hour. Remember to sign the honor code on the last page. Good luck!

Problem 1 (True-false, 10 points)**.** You will get 1 point for a correct answer and 0 points for blank and incorrect answers.[1] No explanations are required.

(1)	There is a unique value of k which makes the vector $(k, 1)$ orthogonal to $(3, 1)$ in $\mathbb{R}^2$.	**T**	F
(2)	A linear system with more variables than equations cannot have exactly one solution.	**T**	F
(3)	The vector $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ is *not* an element of the set $\left\{ x \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + y \begin{pmatrix} -3 \\ -1 \\ 1 \end{pmatrix} : x, y \in \mathbb{R} \right\}$.	**T**	F
(4)	There is a unique function $f : \mathbb{R} \to \mathbb{R}$ of the form $f(x) = ax^2 + bx + c$ whose graph passes through the three points $(1, 2), (-1, 6), (0, 3)$.	**T**	F
(5)	The set $$\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x - y + 3z = 0 \right\}$$ is a vector space (under the usual operations inherited from $\mathbb{R}^3$.)	**T**	F
(6)	The set $\{p(x) \in \mathcal{P}_2 : p(1) = 0\}$ is a vector space (under the usual operations inherited from $\mathcal{P}_2$).	**T**	F
(7)	The set of polynomials of degree at least 3, together with the zero polynomial, is a vector space (under the usual operations inherited from $\mathcal{P}$).	T	**F**
(8)	The set $$\left\{ \begin{pmatrix} a & 1 \\ 0 & b \end{pmatrix} \in \mathcal{M}_{2\times 2} : a, b \in \mathbb{R} \right\}$$ is a vector space (under the usual operations inherited from $\mathcal{M}_{2\times 2}$).	T	**F**
(9)	If $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$ are vectors such that $\vec{u}$ is orthogonal to $\vec{v}$ and $\vec{v}$ is orthogonal to $\vec{w}$, then $\vec{u}$ must be orthogonal to $\vec{w}$.	T	**F**
(10)	The set of all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that $\int_0^1 f(x)\,dx = 0$ is a vector space (under the usual operations inherited from the vector space of all functions).	**T**	F

[1] During quiz revisions, you'll have the chance to meet with me one-on-one and convince me that you understand and can fully explain *up to 2 of the true-false questions that you left blank* to get full credit on those questions.

Problem 2 (3 points, 1 point each)**.** For each of the following linear systems, indicate how many solutions the system has. You don't need to say what the solutions are, if there are any; you only need to say *how many* there are.

(a) $\begin{cases} x + y = 4 \\ y - z = 0 \end{cases}$

Solution. The system is already in echelon form and z is a free variable, so there are infinitely many solutions.

(b) $\begin{cases} x - y = 3 \\ 2x - 2y = 3 \end{cases}$

Solution. The equations are inconsistent, so there are no solutions.

(c) $\begin{cases} x - y = 3 \\ x + y = 3 \end{cases}$

Solution. Doing just one step of row reduction, we find that there is a leading entry in every non-augmented column.

$$\left(\begin{array}{cc|c} 1 & -1 & 3 \\ 1 & 1 & 3 \end{array}\right) \to \left(\begin{array}{cc|c} 1 & -1 & 3 \\ 0 & 2 & 0 \end{array}\right)$$

Thus there is a unique solution.

Problem 3 (2 points)**.** Suppose that you are trying to solve a system of 2 equations in the 3 unknowns x, y, and z. After row reducing, you obtain the matrix

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -2 \end{array}\right).$$

Describe the solution set in parametric form using vectors, and use this parametrization to explain what the solution set looks like geometrically (write a clear sentence, or draw a clear picture, or both).

Solution. The reduced row echelon form tells us that solution vectors (x, y, z) satisfy

$$x = 1 \text{ and } y + z = -2.$$

In other words, solutions are vectors of the form

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ -2 - z \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} + z \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

Geometrically, the solution set is the line passing through the point $(1, -2, 0)$ in the direction of the vector $(0, -1, 1)$. In other words, this is the unique line passing through the two points $(1, -2, 0)$ and $(1, -3, 1)$.

Problem 4 (3 points)**.** Sunny tells you that he has a piggy bank holding pennies, nickels and dimes that contains 12 coins with a total value of 92 cents. Is this possible? If so, how many coins of each type are in the box, and is this the only possibility? If it is not possible, why not? Explain your answer fully.

Solution. Let x, y, z be the numbers of pennies, nickels, and dimes, respectively. The problem tells us that

$$x + y + z = 12$$
$$x + 5y + 10z = 92.$$

Row reducing to solve this system, we find that z is a free variable and

$$x = -8 + \frac{5}{4}z \text{ and } y = 20 - \frac{9}{4}z.$$

We want to find nonnegative integers x, y, z which satisfy these equations.

Note that z must be between 0 and 9. One option is that we could test all the possible integer values of z from 0 to 9 and find the value(s) of z that makes x and y also be nonnegative integers, if there is any such value of z. It will turn out that $z = 8$ is the only such value.

Alternatively, we can be lazy and use a bit of mathematical reasoning so that we have to do fewer calculations. In order for x and y to also be integers, we see that z must be divisible by 4 (otherwise the fractions won't go away). In order for x to be nonnegative, we need $z \geq 32/5 = 6.4$. Thus the only possibility is $z = 8$. Plugging into the above equations, we find $x = 2$ and $y = 2$, which are in fact both nonnegative integers, so this is a valid solution.

So it is possible for Sunny to be telling the truth; and, if he is telling the truth, the only possibility is that his piggy bank has 2 pennies, 2 nickels, and 8 dimes.

Problem 5 (Extra Credit! 2 points, no partial credit)**.** Let us say that *a Fibonacci sequence* is an infinite list of real numbers $(a_0, a_1, a_2, a_3, \dots)$ where $a_{n+2} = a_{n+1} + a_n$ for all $n \geq 0$. For example, the infinite list

$$(1, 1, 2, 3, 5, 8, \dots)$$

is one example of a Fibonacci sequence.

Consider the set S of all Fibonacci sequences. Is S a vector space? If so, describe what the vector addition and scalar multiplication operations are, explain why the outputs of these operations are Fibonacci sequences, and say what the zero vector is. If not, explain why S fails to be a vector space.

Solution. It is a vector space. Given two Fibonacci sequences $(a_0, a_1, \dots)$ and $(b_0, b_1, \dots)$, we can add them componentwise: in other words, we define

$$(a_0, a_1, \dots) + (b_0, b_1, \dots) = (a_0 + b_0, a_1 + b_1, \dots).$$

This is a Fibonacci sequence because, if c_n denotes the nth term of the sum, then

$$c_{n+2} = a_{n+2} + b_{n+2} = (a_{n+1} + a_n) + (b_{n+1} + b_n) = (a_{n+1} + b_{n+1}) + (a_n + b_n) = c_{n+1} + c_n,$$

for all $n \geq 0$. Similarly, we define scalar multiplicaton componentwise also. In other words, if $\lambda \in \mathbb{R}$ is a scalar, we set

$$\lambda \cdot (a_0, a_1, \dots) = (\lambda a_0, \lambda a_1, \dots).$$

If c_n denotes the nth term of this product, then

$$c_{n+2} = \lambda a_{n+2} = \lambda(a_{n+1} + a_n) = \lambda a_{n+1} + \lambda a_n = c_{n+1} + c_n$$

for all $n \geq 0$, so the product is again a Fibonacci sequence. The zero vector is the sequence $(0, 0, 0, \dots)$ where all terms are 0.

Honor code. If you have neither given nor received any unauthorized aid on this quiz, please write either "HCU" or "Honor Code Upheld" below, and sign your name next to it.

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